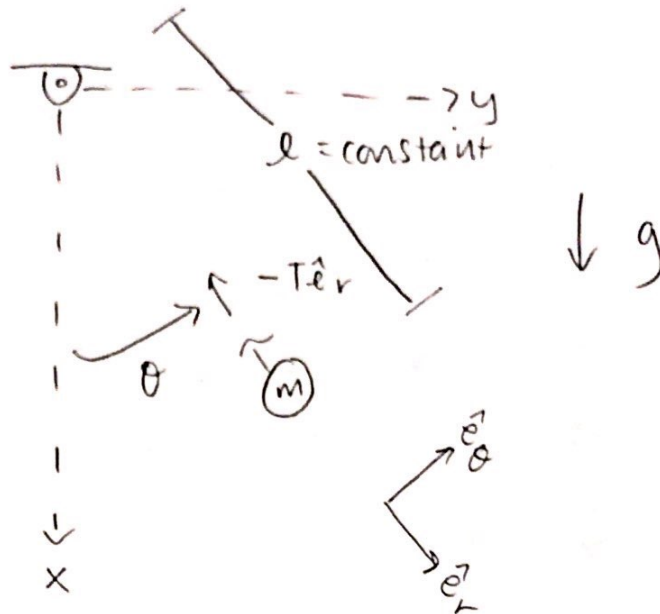
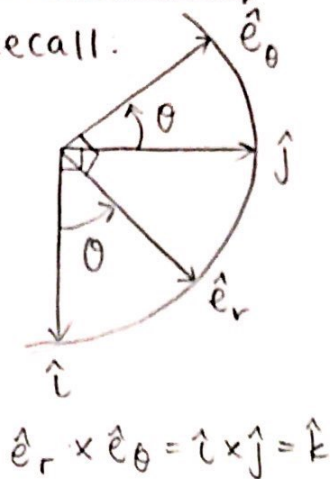


Recall:



$\hat{i}, \hat{j}, \hat{e}_r, \hat{e}_\theta$ or $\times \hat{i}, \hat{j}, \hat{e}_r, \hat{e}_\theta = \pm 1$ or $\pm \cos \theta$ or $\pm \sin \theta$
or 0 (times, maybe, $\hat{k}$)

Big question: find EoM?

Last class, several ways

Method 4 Power Balance

$$P = \dot{E}_k$$

$$P_{\text{string}} + P_{\text{gravity}} = \frac{1}{2} m v^2$$

$-T \hat{e}_r \cdot \vec{v} = 0$

$$\boxed{\hat{i} \cdot \hat{e}_\theta = -\sin \theta} \quad \underbrace{mg \hat{i}}_F \cdot \underbrace{l \dot{\theta} \hat{e}_\theta}_{\vec{v}} = \frac{d}{dt} \left(\frac{1}{2} m (l \dot{\theta})^2 \right)$$

$\Rightarrow$ Algebra

$$\boxed{\ddot{\theta} = -\frac{g}{l} \sin \theta}$$

Method 5 Lagrange Equations

• Minimal coordinates: θ

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}m\ell^2\dot{\theta}^2$$

$$E_p = mgh = -mg\ell\cos\theta \quad (\text{define as } 0 \text{ halfway up circle})$$

$$\mathcal{L} = E_k - E_p$$

$$\mathcal{L}(\theta, \dot{\theta}) = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell\cos\theta$$

L.E.s:

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\frac{d}{dt}(m\ell^2\dot{\theta}) - (-mg\ell\sin\theta) = 0$$

$$\Rightarrow \boxed{\ddot{\theta} = -\frac{g}{\ell}\sin\theta}$$

Method 6 DAE 1

$$\vec{F} = m\vec{a} ; -T\hat{e}_r + mg\hat{e}_t = m(\ddot{x}\hat{i} + \ddot{y}\hat{j})$$

"maximal" coord: x, y

Constraint eqns:

$$x^2 + y^2 = \ell^2 = \text{constant} \quad (*)$$

$$\frac{d}{dt} (*) \Rightarrow x\dot{x} + y\dot{y} = 0 \quad (1)$$

$$\frac{d}{dt} (1) \Rightarrow \dot{x}^2 + x\ddot{x} + \dot{y}^2 + y\ddot{y} = 0$$

$$x\ddot{x} + y\ddot{y} = -\dot{x}^2 - \dot{y}^2 \quad (2)$$

$$\boxed{\text{r.h.s file}} \rightarrow \boxed{\text{state}} = \dot{x}, \ddot{x}, \dot{y}, \ddot{y}$$

$$\uparrow$$
$$\text{state} = x, \dot{x}, y, \dot{y}$$

$$\begin{aligned}
 \hat{i} \cdot \{\vec{F} = m\vec{a}\} &\Rightarrow m \cdot \ddot{x} + 0 \cdot \ddot{y} + \cos\theta T = mg \\
 \hat{j} \cdot \{\vec{F} = m\vec{a}\} &\Rightarrow 0 \cdot \ddot{x} + m \cdot \ddot{y} + \sin\theta T = 0 \\
 \text{constraint} &\Rightarrow x \dot{x} + y \dot{y} + 0 T = -\dot{x}^2 - \dot{y}^2
 \end{aligned}
 \left. \vphantom{\begin{aligned} \hat{i} \cdot \{\vec{F} = m\vec{a}\} \\ \hat{j} \cdot \{\vec{F} = m\vec{a}\} \\ \text{constraint} \end{aligned}} \right\} \begin{array}{l} \text{linear in} \\ \dot{x}, \dot{y}, T \end{array}$$

-replace $\cos\theta$ w/ $\frac{x}{\sqrt{x^2+y^2}}$, $\sin\theta$ w/ $\frac{y}{\sqrt{x^2+y^2}}$

$$\Rightarrow \underbrace{\begin{bmatrix} m & 0 & \frac{x}{\sqrt{x^2+y^2}} \\ 0 & m & \frac{y}{\sqrt{x^2+y^2}} \\ x & y & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ T \end{bmatrix}}_W = \underbrace{\begin{bmatrix} mg \\ 0 \\ -\dot{x}^2 - \dot{y}^2 \end{bmatrix}}_b$$

$$W = A \setminus b$$

$$\ddot{x} = w(1)$$

$$\ddot{y} = w(2)$$

Note:

- As errors grow, errors in constraint also grow
- Need consistent IC's - need to satisfy (*) and (1)

$$y_0^2 + x_0^2 = l^2 \quad (\text{from } (*))$$

$$x_0 \dot{x}_0 + y_0 \dot{y}_0 = 0 \quad (\text{from } (1))$$

Analogy:

We want $x=0$ for all t

$$\Leftrightarrow x(0)=0, \dot{x}(0)=0, \ddot{x}(0)=0 \text{ for all } t$$

Method 7 DAE 2

$$\vec{F} = m\vec{a} \Rightarrow mg\hat{e}_r - T\hat{e}_r = m[(\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta]$$

constraint equation: $r = l \Rightarrow \dot{r} = 0 \Rightarrow \ddot{r} = 0$

$$\begin{bmatrix} \{\vec{F} = m\vec{a}\} \cdot \hat{e}_r \\ \{\vec{F} = m\vec{a}\} \cdot \hat{e}_\theta \\ \text{constraint eqn} \end{bmatrix} = \begin{bmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \end{bmatrix} \begin{bmatrix} \ddot{r} \\ \ddot{\theta} \\ T \end{bmatrix} = \begin{bmatrix} \square \\ \square \\ \square \end{bmatrix} \leftarrow b$$

$\uparrow$ A $\uparrow$ terms involving $r, \dot{r}, \theta, \dot{\theta}$ parameters

$$\begin{bmatrix} \ddot{r} \\ \ddot{\theta} \\ T \end{bmatrix} = A \setminus b$$